Time (UTC or Mission Elapsed Time)	Robot Function / Mode	Primary Objective / Milestone	Operation Mode (Teleop / Autonomous / Semi-Autonomous / Scripted)	Inputs (e.g., command, signal, state)	Expected Outputs (e.g., telemetry, sensor data)	Data Products Generated	Verification Method	Associated Risk or Sensitivity (High/Med/Low)	Dependencies / Predecessor Actions	Estimated Duration	Notes
Launch Day	Launch & Commission	Deploy Spacecraft And Establish Communications	Scripted/Autonomous	Launch separation signal, startup sequences	Telemetry, power status, comm link	Initial health report	Verify separation and subsystem health	High	Launch vehicle deployment	1 day	Mission Start
Cruise Years 1-6	Cruise & Approach	Transit To Europa	Semi-Autonomous	Navigation measurements, trajectory updates	Navigation telemetry	Trajectory correction logs	Verify trajectory tolerance	Medium	Launch Complete	6 years	Periodic health monitoring
Europa Arrival	Europa Orbit Insertion	Achieve Stable Europa Orbit	Autonomous	Orbital state measurements	Orbital telemetry	Orbit determination solution	Verify stable orbit	High	Cruise Complete	1 day	Critical maneuver
Orbit Day 1-15	Terrain Mapping	Generate Landing Candidates	Autonomous	Orbital imagery, radar scans	Hazard maps	Global terrain maps	Verify terrain coverage	Medium	Orbit Insertion complete	15 das	Orbital Agent 1
Orbit Day 15-30	Communications Relay & Localization Support	Validate Landing Regions	Autonomous	Orbital navigation data	Localization updates	Navigation support database	Verify location accuracy	Medium	Terrain Mapping complete	15 days	Orbital Agent 2
Surface Day 0	Rover Landing	Safe Landing	Autonomous	Landing guidance sensors	Landing telemetry	Landing performance report	Verify touchdown location	High	Landing Site selected	4 hrs	Highest risk phase
Surface Day 1	Payload Initialization	Activate Science Payloads	Autonomous	Startup commands	Instrument health telemetry	Payload status report	Verify payload functionality	Medium	Landing Complete	6 hrs	New 2.4 activity
Surface Day 1-2	Radar Calibration	Calibrate Radar Sonar	Autonomous	Calibration routines	Calibration measurements	Calibration dataset	Verify measurement accuracy	Low	Payload initialization complete	1 day	New 2.4 activity
Surface Day 2-5	Traverse Planning	Generate Survey Route	Autonomous	Terrain maps, hazard maps	Planned route	Navigation plan	Verify route safety	Medium	Calibration complete	3 days	New 2.4 activity
Surface Day 5-20	Autonomous Navigation	Transverse Survey Area	Autonomous	Terrain maps, localization estimates	Position updates	Traverse logs	Verify localization accuracy	Medium	Route planning complete	15 days	Expanded from 2.3
Surface Day 5-20	Radar Data Collected	Acquire Subsurface Measurements	Autonomous	Radar returns	Ice thickness estimates	Radar profiles	Verify coverage requirements	Medium	Navigation Active	15 days	Expanded from 2.3
Surface Day 10-22	Subsurface Map Generation	Create Ice-Shell Model	Autonomous	Radar profiles	Subsurface maps	Ice thickness map	Verify map completeness	Medium	Data collection underway	12 days	Expanded from 2.3
Surface Day 22-26	Consensus Validation	Rank Candidate Cryobot Spots	Autonomous	Terrain maps, Ice maps	Site rankings	Site evaluation report	Verify selection criteria	Low	Map generation complete	4 days	New 2.4 activity
Surface Day 26-28	Cryobot Site selection	Select Optimal Insertion Site	Autonomous	Candidate rankings	Selected coordinates	Cryobot insertion package	Verify mission criteria	Low	Consensus validation complete	2 days	Mission success metric
Surface Day 29-30	Data transmission & End Mission	Downlink Archive And Close Mission	Semi-Autonomous	Science archive	Archive transmission	Final mission report	Verify archive receipt	Low	Site selection complete	1 day	Mission end